

Curriculum Vitae

Carlos Ignacio Hernández Castellanos

October 5, 2026

1 Personal data

Name:	Carlos Ignacio Hernández Castellanos
E-mail:	carlos.hernandez@iimas.unam.mx
Highest university diploma:	Ph. D. on Computer Science
Current place of employment:	National Autonomous University of Mexico
Current position:	Associate Researcher, Full-time, Level C
PRIDE:	Level C (from July 2026)
Telephone:	+52 55 5622 3899 ext 44213

1.1 Academic background

- Ph. D. in Computer Science
CINVESTAV Computer Science Department
Thesis: Set Oriented Methods for Multi-objective Optimization
Adviser: Oliver Schütze
2013-2017
- M. Sc. in Computer Science
CINVESTAV Computer Science Department
Thesis: Cell-to-Cell Mapping for Global Multi-Objective Optimization
Adviser: Oliver Schütze
2011-2013
- B. Sc. in Computer Systems Engineering
Technological Institute of Tepic
2007-2011

1.2 Profesional Experience

- Head of B. Sc. on Data Science
Institute for Research on Applied Mathematics and Systems of the National Autonomous University of Mexico (UNAM)
2026-present
- Associate Researcher
Institute for Research on Applied Mathematics and Systems of the National Autonomous University of Mexico (UNAM)
2021-present

- Postdoctoral Researcher
CINVESTAV-IPN, Mexico
2020
- Postdoctoral Researcher
University of Oxford, UK
2018-2019
- Chief Technology Officer and Co-founder
Mutuo Financiera, Mexico
2017-2021

2 Research projects

- Principal Researcher
A New Algorithmic Framework for Multi-Objective Reinforcement Learning (IA102025)
National Autonomous University of Mexico.
- Co-principal researcher
Google Academic Research Award 2024 on Society-Centered AI
Community weavers of artificial intelligence and nature-based solutions.
A grassroots-participatory AI approach to urban transformations towards sustainability
National Autonomous University of Mexico
- Principal Researcher
Development of Multi-objective Optimization Algorithms for Highly Complex Environments (2023-2024)
National Autonomous University of Mexico
- Postdoctoral Researcher
Multi-objective optimization under uncertainty with applications to optimal control (November 2018 - October 2019)
PI: Dr. Sina Ober-Blöbaum
University of Oxford
- Ph. D. student
Set Oriented Methods for Multi-objective Optimization (September 2013 - 2017)
Advisers: Dr. Oliver Schütze
CINVESTAV Department of Computer Science
- Project member
Design of Test Functions for Multi-objective Optimization in Noisy Environments (October 2015)
Advisers: Dr. Oliver Schütze and Dr. Peter Fleming
Sheffield University Department of Mechanical Engineering (1 month stay)
- Project member
Generalized Cell Mapping for Multi-objective Optimization (June - July 2014)

Advisers: Dr. Oliver Schütze and Dr. Qiao Ding
Tianjin University Department of Mechanical Engineering (1 month stay)

3 Industry Collaboration Projects

1. Specification Definition for a Strategic Roadmap: consulting activities with a focus on data science and optimization
Participant
2025
2. Analysis of the Optimization Model (DGAJ-DPI-40-050324-101)
Project Lead
2024
3. Feasibility Assessment for the Application of Artificial Intelligence in Processes (DGE-GDRP-ONE-02-2024)
Participant
2024

4 Publications

Monographs

1. Schütze, O. and Hernández, C. (2021). Archiving Strategies for Multi-objective Evolutionary Optimization Algorithms. Series: Nonlinear Systems and Complexity. Springer International Publishing.
2. Sun, J.-Q., Xiong, F.-R., Schütze, O., Hernández, C. (2018). “Cell Mapping Methods - An Algorithmic Approach and Applications.” Series: Nonlinear Systems and Complexity. Springer International Publishing.

JCR Journals

1. Cassis, J. P., Gatica, J., Rojas-Morales, N., and Hernández Castellanos, C. (2026). MHMetLib: A New Performance Indicators Library for Trajectory and Population-Based Metaheuristics. Applied Sciences, 16(7), 3529.
2. Hernández, C., Rodriguez-Fernandez, A. E., Schäpermeier, L., Cuate, O., Trautmann, H., and Schütze, O. (2025). An Evolutionary Approach for the Computation of ϵ -Locally Optimal Solutions for Multi-Objective Multimodal Optimization. IEEE Transactions on Evolutionary Computation.
3. Rivera, J. A. L., Castellanos, C. I. H., Morales, R., and Cortés, L. A. P. (2025). On the multi-objective hyperparameter optimization of the weighted entropic associative memory. Scientific Reports, 15(1), 38772.
4. García-Núñez, D., Rodríguez-Vázquez, K., Hernández, C., and Galván, E. (2025). Neuroevolution-based multiobjective algorithm for feature selection and binary classification of DNA microarrays. Applied Soft Computing, 113520.

5. Galindo-Hernández, R., Rodríguez-Vázquez, K., Galán-Vásquez, E., and Hernández Castellanos, C. I. (2025). Online-adjusted evolutionary biclustering algorithm to identify significant modules in gene expression data. *Briefings in Bioinformatics*, 26(1), bbae681.
6. O. Schütze, A. E. Rodríguez-Fernandez, C. Segura and C. Hernández (2025). Finding the Set of Nearly Optimal Solutions of a Multiobjective Optimization Problem. *IEEE Transactions on Evolutionary Computation*, vol. 29, no. 1, pp. 145-157.
7. Lara Moreno, Y., and Hernández Castellanos, C. I. (2024). A Hierarchical Approach to a Tri-Objective Portfolio Optimization Problem Considering an ESG Index. *Mathematics*, 12(19), 3145.
8. A. E. Rodriguez-Fernandez, L. Schäpermeier, C. Hernández, P. Kerschke, H. Trautmann and O. Schütze (2024). Finding ϵ -Locally Optimal Solutions for Multi-Objective Multimodal Optimization, in *IEEE Transactions on Evolutionary Computation*, doi: 10.1109/TEVC.2024.3458855.
9. López-Ruiz, S., Hernández-Castellanos, C. I., and Rodríguez-Vázquez, K. (2024). Multi-objective optimization of neural network with stochastic directed search. *Expert Systems with Applications*, 237, 121535.
10. Enríquez-Zárate, J., Gómez-Peñate, S., Hernández, C., Villarreal-Valderrama, F., Velázquez, R., and Trujillo, L. (2023). Optimization of vibration control using a hybrid scheme with sliding-mode and positive position feedback. *Optimal Control Applications and Methods*.
11. Hernandez Castellanos, C. I., Ober-Blöbaum, S., and Peitz, S. (2020). Explicit multiobjective model predictive control for nonlinear systems under uncertainty. *International Journal of Robust and Nonlinear Control*, 30(17), 7593-7618.
12. Hernández Castellanos, C. I., Schütze, O., Sun, J. Q., Morales-Luna, G., and Ober-Blöbaum, S. (2020). Numerical Computation of Lightly Multi-Objective Robust Optimal Solutions by Means of Generalized Cell Mapping. *Mathematics*, 8(11), 1959.
13. Schuetze, O., Hernandez, C., Talbi, E. G., Sun, J. Q., Naranjani, Y., and Xiong, F. R. (2019). Archivers for the representation of the set of approximate solutions for MOPs. *Journal of Heuristics*, 25(1), 71-105..
14. Sardahi, Y., Sun, J. Q., Hernández, C., Schütze, O. (2017). Many-Objective Optimal and Robust Design of Proportional-Integral-Derivative Controls With a State Observer. *Journal of Dynamic Systems, Measurement, and Control*, 139(2).
15. Qin, Z. C., Xiong, F. R., Ding, Q., Hernández, C., Fernandez, J., Schütze, O., Sun, J. Q. (2017). Multi-objective optimal design of sliding mode control with parallel simple cell mapping method. *Journal of Vibration and Control*, 23(1), 46-54.

16. Fernández, J., Schütze, O., Hernández, C., Sun, J. Q., Xiong, F. R. (2016). Parallel simple cell mapping for multi-objective optimization. *Engineering Optimization*, 48(11), 1845-1868.
17. Xiong, F. R., Qin, Z. C., Ding, Q., Hernández, C., Fernández, J., Schütze, O., Sun, J. Q. (2015). Parallel cell mapping method for global analysis of high-dimensional nonlinear dynamical systems. *Journal of Applied Mechanics*, 82(11), 111010.

Other journals

1. Hernández Castellanos, C. I., and Schütze, O. (2022). A Bounded Archiver for Hausdorff Approximations of the Pareto Front for Multi-Objective Evolutionary Algorithms. *Mathematical and Computational Applications*, 27(3), 48.
2. Khan, M. A., Sardahi, Y., and Hernández Castellanos, C. I. (2022). Multi-Objective and Robust Design of a Semi-Active Suspension System. *International Journal on Engineering, Science and Technology*, 4(1), 1-13.
3. Hernández Castellanos, C. I., Schütze, O., Sun, J. Q., and Ober-Blöbaum, S. (2020). Non-Epsilon Dominated Evolutionary Algorithm for the Set of Approximate Solutions. *Mathematical and Computational Applications*, 25(1), 3.
4. Enríquez-Zárate, J., Trujillo, L., Toledo-Ramírez, G.K., Ramos-Cirilo, Á.J., Hernández, C. (2018). "Optimization of PPF control of a building-like structure for vibration control." *Computacion y Sistemas*, 22 (2), pp. 413-424.
5. Naranjani, Y., Hernández, C., Xiong, F.R., Schütze, O., Sun, J.-Q. (2016). "A hybrid method of evolutionary algorithm and simple cell mapping for multi-objective optimization problems." *International Journal of Dynamics and Control*.
6. Hernández, C., Naranjani, Y., Sardahi, Y., Liang, W., Schütze, O., Sun, J. Q. (2013). "Simple cell mapping method for multi-objective optimal feedback control design." *International Journal of Dynamics and Control*, 1(3), 231-238.
7. Xiong, F., Qin, Z., Hernández, C., Sardahi, Y., Naranjani, Y., Liang, W., Sun, J. (2013). "A multi-objective optimal PID control for a nonlinear system with time delay." *Theoretical and Applied Mechanics Letters*, 3(6).

Book chapters

1. Enríquez-Zárate, J., Trujillo, L., Hernández, C., and Sánchez, C. N. (2017, September). Optimal Design of Sliding Mode Control Combined with Positive Position Feedback. In *Numerical and Evolutionary Optimization* (pp. 187-199). Springer, Cham.
2. Hernández, C., Schütze, O., Sun, J. Q. (2017). Global Multi-objective Optimization by Means of Cell Mapping Techniques. In *EVOLVE—A Bridge between Probability, Set Oriented Numerics and Evolutionary Computation VII* (pp. 25-56). Springer International Publishing.

In proceedings of international conference

1. Rodríguez-Fernandez, A. E., Hernández Castellanos, C. I., and Schütze, O. (2025, February). On the Approximation of the Entire Pareto Front of a Constrained Multi-objective Optimization Problem. In International Conference on Evolutionary Multi-Criterion Optimization (pp. 85-98). Singapore: Springer Nature Singapore.
2. López-Ruiz, S., Hernández-Castellanos, C. I., and Rodríguez-Vázquez, K. (2022, July). Multi-objective framework for quantile forecasting in financial time series using transformers. In Proceedings of the Genetic and Evolutionary Computation Conference (pp. 395-403).
3. Hernández, C., and Schütze, O. (2022, July). A Bounded Archive Based for Bi-objective Problems based on Distance and epsilon-dominance to avoid Cyclic Behavior. In Proceedings of the Genetic and Evolutionary Computation Conference (GECCO-2022), Boston, MA, USA (pp. 9-13).
4. Khan, M. A., Sardahi, Y., and Hernández Castellanos, C. I. (2021). Robust Multi-Objective Optimal Design of a Racing Car Suspension System. In Proceedings of International Conference on Engineering, Science and Technology (p. 12).
5. Wang, H., Hernández Castellanos, C. I., and Eftimov, T. (2021, March). On Statistical Analysis of MOEAs with Multiple Performance Indicators. In International Conference on Evolutionary Multi-Criterion Optimization (pp. 26-37). Springer, Cham.
6. Menchaca-Mendez, A., Hernández, C., Coello, C. A. C. (2016). “ Δ_p -MOEA: A new multi-objective evolutionary algorithm based on the Δ_p indicator.” En Evolutionary Computation (CEC), 2016 IEEE Congress on (pp. 3753-3760). IEEE.
7. Qin, Z. C., Xiong, F. R., Ding, Q., Hernández, C., Fernandez, J., Schütze, O., Sun, J. Q. (2015, August). “Multi-Objective Optimal Design and Validation of Sliding Mode Control.” In ASME 2015 International Design Engineering Technical Conferences and Computers and Information in Engineering Conference (pp. V008T13A015-V008T13A015). American Society of Mechanical Engineers.
8. P. Kerschke, M. Preuss, Hernández, C., Schütze, O., Sun, J.-Q., C. Grimme, G. Rudolph, B. Bischl and H. Trautmann (2014). “Cell Mapping Techniques for Exploratory Landscape Analysis.” Proceedings of EVOLVE 2014 - A Bridge between Probability, Set Oriented Numerics, and Evolutionary Computing. Beijing, P. R. China.
9. Hernández, C., Schütze, O., Emmerich, M., Xiong, F.-R. and Sun, J.-Q. (2014). “Barrier Tree for Continuous Landscapes by Means of Generalized Cell Mapping.” Proceedings of EVOLVE 2014 - A Bridge between Probability, Set Oriented Numerics, and Evolutionary Computing. Beijing, P. R. China.

10. Hernández, C., Sun, J.-Q. and Schütze, O. (2013). “Computing the Set of Approximate Solutions of a Multi-Objective Optimization Problem by Means of Cell Mapping Techniques.” Proceedings of EVOLVE 2013 - A Bridge between Probability, Set Oriented Numerics, and Evolutionary Computing. Leiden, The Netherlands.
11. Naranjani, Y., Hernández, C., Xiong, F.-R., Schütze, O. and Sun, J.-Q. (2013). “A Hybrid Algorithm for the Simple Cell Mapping Method in Multi-objective Optimization.” Proceedings of EVOLVE 2013 - A Bridge between Probability, Set Oriented Numerics, and Evolutionary Computing. Leiden, The Netherlands.
12. Naranjani, Y., Y. Sardahi, Sun, J.-Q., Hernández, C. and Schütze, O. (2013). “Fine Structure of Pareto Front of Multi-objective Optimal Feedback Control Design.” Proceedings of the 2013 ASME Dynamic Systems and Control Conference - Control, Monitoring, and Energy Harvesting of Vibratory Systems. Stanford, California. DSCC2013-3944.
13. Y. Sardahi, Naranjani, Y., W. Liang, Sun, J.-Q., Hernández, C. and Schütze, O. (2013). “Multi-objective Optimal Control Design with the Simple Cell Mapping Method.” Proceedings of ASME 2013 International Mechanical Engineering Congress & Exposition. San Diego, California. IMECE2013-65506.
14. Y. Sardahi, Y. Narajani, W. Liang, Xiong, F.-R., Z.-C. Qin, Y. Xue, Hernández, C., Schütze, O., Sun, J.-Q. (2013). “Multi-objective Optimal Design of Feedback Controls for Dynamical Systems Time Delay.” Proceeding of CSTAM CONFERENCE-2013.

4.1 Extended Abstract in an International Conference

1. Becerril Torres, T., Hernández Castellanos, C. I., and Santana Hermida, R. (2025). Particle Swarm Algorithm for Multi-Objective Reinforcement Learning. In Multi-Objective Decision Making Workshop at ECAI 2025.
2. Millán, A. M., and Hernández Castellanos, C. I. (2025, July). Uncertainty Quantification of the Hypervolume for Evolutionary Multi-Objective Reinforcement Learning. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (pp. 399–402).
3. Santana Hermida, R., and Hernández Castellanos, C. I. (2025, July). Benchmarking MOEAs for Solving Continuous Multi-Objective Reinforcement Learning Problems. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (pp. 415–418).
4. Borrel Miller, S. and Hernández Castellanos, C. (2024, July). A R2-Based Multi-Objective Reinforcement Learning Algorithm. LION 18.
5. Velázquez Cruz, R. F., and Hernández Castellanos, C. I. (2024, July). A Cooperative Multi-Indicator-Based Ant Colony Optimization Algorithm for the MOGenConVRP. In Proceedings of the Genetic and Evolutionary Computation Conference Companion (pp. 215–218).

6. Espinoza, Ó., Rodríguez-Vázquez, K., Hernández, C. I., and Rodríguez-Romo, S. (2023, July). Comparison of Three Versions of the Whale Optimization Algorithm (WOA) on the BBOB Test Suite. In *Proceedings of the Companion Conference on Genetic and Evolutionary Computation* (pp. 1595–1602).
7. Sanchez Sanchez, V. M., Gershenson Garcia, C., and Hernandez Castellanos, C. I. (2023, July). On the Effect of Temporal Heterogeneity on Selection Pressure of Evolutionary Algorithms. In *Proceedings of the Companion Conference on Genetic and Evolutionary Computation* (pp. 479–482).
8. Hernández, C., and Schütze, O. (2022, July). Archivers for Single- and Multi-Objective Evolutionary Optimization Algorithms. In *Proceedings of the Genetic and Evolutionary Computation Conference Companion* (pp. 37–38).
9. García-Núñez, D., Rodríguez-Vázquez, K., and Hernández, C. (2022, July). Neuroevolution-Based Multi-Objective Algorithm for Gene Selection and Microarray Classification. In *Proceedings of the Genetic and Evolutionary Computation Conference Companion* (pp. 647–650).
10. López, E. M., and Hernández Castellanos, C. I. (2020, July). A Local Hypervolume Contribution Scheme to Improve Spread of the Pareto Front and Computational Time. In *Proceedings of the 2020 Genetic and Evolutionary Computation Conference Companion* (pp. 167–168).
11. Hernández Castellanos, C. I., Ober-Blöbaum, S., and Peitz, S. (2021). Explicit Multi-Objective Model Predictive Control for Nonlinear Systems Under Uncertainty. *31st European Conference on Operational Research*, July 2021.
12. Hernández Castellanos, C. I., Ober-Blöbaum, S., and Peitz, S. (2021). Explicit Multi-Objective Model Predictive Control for Nonlinear Systems Under Uncertainty. *The 22nd Conference of the International Federation of Operational Research Societies*, August 2021.
13. López, E. M., and Hernández Castellanos, C. I. (2020, July). A Local Hypervolume Contribution Scheme to Improve Spread of the Pareto Front and Computational Time. In *Proceedings of the 2020 Genetic and Evolutionary Computation Conference Companion* (pp. 167–168).
14. Hernández, C. and Ober-Blöbaum, S. (2019). “A Decomposition-Based EMOA for Set-Based Robustness.” *The Genetic and Evolutionary Computation Conference 2019*.
15. Hernández, C., and Ober-Blöbaum, S. (2019). “Reference Point Methods for Set-Based Robustness by Means of the Hausdorff Distance.” *30th European Conference on Operational Research*.
16. Hernández, C., and Ober-Blöbaum, S. (2019). “A Differential Evolution Reference Point Method for Set-Based Robustness by Means of the Averaged Hausdorff Distance.” In *25th International Conference on Multiple Criteria Decision Making*.

17. Hernández, C., Schütze, O., Emmerich, M., Xiong, F.-R. and Sun, J.-Q. (2014). “Barrier Tree for Continuous Landscapes by Means of Generalized Cell Mapping.” Proceedings of EVOLVE 2014 – A Bridge between Probability, Set Oriented Numerics, and Evolutionary Computing. Beijing, P. R. China.

5 Awards

- Afirme-UNAM Award 2025 in the research category.
- Google Academic Research Award 2024 in AI centric society.
- National Researcher Level I in the period 2019-2030.
- Rosenblueth award to best Ph. D. thesis in technology and engineering (2018) by CINVESTAV-IPN.
- Nayarit State Youth Award (Premio Estatal de la Juventud Nayarit) 2018 in academic achievement. This award is granted by the government of the State of Nayarit.
- 3rd place in Best Ph. D. Thesis Award on Artificial Intelligence (2019). This award is granted by the Mexican Society of Artificial Intelligence (SMIA).
- Invited to the Dagstuhl Seminar, Germany (2016), on automated algorithm selection and configuration.
- 1st place in Best Master’s Thesis Award on Artificial Intelligence (2015). This award is granted by the Mexican Society of Artificial Intelligence (SMIA).

6 Citations

According to Google Scholar https://scholar.google.com/citations?hl=es&user=nMMBg2gAAAAJ&view_op=list_works&sortby=pubdate (seen on October 5th, 2026):

- Citations: 787
- H index: 15
- i10 index: 21

7 Students

7.1 Postdoctoral Research

1. Gerardo Altamirano Gómez – Evolutionary design of deep invariance-learning architectures for object recognition. 2024–2025.

7.2 PhD

1. Daniel Alejandro Alonso Bastos – Modeling and navigation of policy landscapes in reinforcement learning using surrogate models based on graph neural networks. Primary advisor, PCIC, ongoing.
2. Rodrigo F. Velázquez Cruz – A robust framework for multi-objective neural architecture search. Primary advisor, PCIC, ongoing.
3. José Olivas Díaz – Multi-objective multi-agent reinforcement learning using evolutionary algorithms. Primary advisor, PCIC, ongoing.
4. Iván Alcalá Paz – Computational alternatives to the perceptron: Kolmogorov–Arnold networks for differential equations. Primary advisor, PCIC, ongoing.
5. Ricardo D. Fernández Noguez – Preference-based multi-objective reinforcement learning. Primary advisor, PCIC, ongoing.
6. José A. Alonso González – Constrained multi-objective reinforcement learning. Primary advisor, PCIC, ongoing.

7.3 Master’s

1. Pablo Uriel Benítez Ramírez – Pareto Tracer-based multi-objective reinforcement learning for multimodal traffic management. Primary advisor, PCIC, ongoing.
2. Fernando R. Valenzuela G. de León – Artificial assistant for choreography creation in dance: an approach from computational creativity. Co-advisor, PCIC, ongoing.
3. Luz Itzel Valdeolivar Hernández – Hyperparameter tuning of the superiorization method through exploratory landscape analysis. Co-advisor, PCIC, ongoing.
4. Baruch Mejía – Speech enhancement through multi-objective optimization in a contextual acoustic representation space. Co-advisor, PCIC, ongoing.
5. Teresa Becerril Torres – Particle swarm algorithm for multi-objective reinforcement learning. Primary advisor, PCIC, August 2026.
6. José A. Alonso González – Multi-objective optimization for the operation of water distribution networks. Co-advisor, PCIC, July 2026.
7. Alberto M. Millán Prado – Uncertainty quantification of the hypervolume in multi-objective reinforcement learning problems. Primary advisor, PCIC, February 2026.
8. María Carmen Aguirre Delgado – When does the weighted sum method work well in multi-task learning? Co-advisor, PCIC, October 2025.
9. José Olivas Díaz – Evaluation of multi-objective multi-agent reinforcement learning algorithms using weighted sums. Primary advisor, PCIC, August 2025.

10. Sofía Borrel Miller – R2-based multi-objective reinforcement learning. Primary advisor, PCIC, June 2025.
11. Rodrigo F. Velázquez Cruz – Multi-indicator ant colony optimization for the multi-objective vehicle routing problem under uncertainty. Primary advisor, PCIC, December 2024.
12. Víctor M. Sánchez Sánchez – Effect of temporal heterogeneity on the selection pressure of evolutionary algorithms. Co-advisor, PCIC, November 2024.
13. Juan A. López Rivera – Parameter optimization in weighted entropic associative memory systems. Co-advisor, PCIC, February 2024.

7.4 Undergraduate

1. Fernando Avitúa Varela – Objective reduction in multi-objective problems using quality indicators. November 2024.

8 Courses

8.1 Graduate

1. Reinforcement Learning - Graduate Program in Computer Science and Engineering, UNAM (2022–2026)
2. Artificial Intelligence - Graduate Program in Computer Science and Engineering, UNAM (2021–2026)
3. Multi-objective Optimization and Multi-criteria Decision-Making - Graduate Program in Computer Science and Engineering, UNAM (2021–2022)
4. Master’s Thesis Seminar II - Master’s Program in Computer Science, Department of Computer Science, CINVESTAV-IPN (2020)

8.2 Undergraduate

1. Pattern Recognition - BSc in Data Science, IIMAS, UNAM (2021–2023)
2. Reinforcement Learning - BSc in Data Science, IIMAS, UNAM (2023)
3. Machine Learning - BSc in Data Science, IIMAS, UNAM (2021, 2025)
4. Evolutionary Computation - Faculty of Science, UNAM (2021)
5. Theory of Computation - BSc in Software Engineering, Autonomous University of Mexico City, Cuautepec Campus (2021)

8.3 Certifications

1. Preparatory Course for Admission to the Applied Artificial Intelligence Diploma Program (5 hours) – Applied Artificial Intelligence Diploma Program, DGTIC, UNAM (2026)
2. Unsupervised Learning (10 hours) – Applied Artificial Intelligence Diploma Program, DGTIC, UNAM (2026)
3. Ethics in Artificial Intelligence (5 hours) – Applied Artificial Intelligence Diploma Program, DGTIC, UNAM (2025–2026)
4. Evolutionary Computation (20 hours) – Applied Artificial Intelligence Diploma Program, DGTIC, UNAM (2024)

9 Outreach and Dissemination

9.1 Coordination

1. AI Summit 2025 - Organizing Committee, Fintual and OpenAI (2025)
2. IIMAS Colloquium 2024 - Organizing Committee, IIMAS, UNAM (2024)

9.2 Participations

1. Hacia la toma de decisiones secuenciales multiobjetivo. 20th Anniversary of the Department of Computer Science, CINVESTAV-IPN, Seminar, 2026-09-25.
2. Una serie de decisiones afortunadas: los algoritmos de elegir bien. Conversa Matemático, IIMAS-UNAM, Talk, 2026-09-24.
3. Cómo la tecnología revoluciona la medicina. Jueves de Ciencia, Fundación UNAM and TV UNAM, Television, 2026-09-24.
4. Cómo cambia la tecnología a la guerra. Jueves de Ciencia, Fundación UNAM and TV UNAM, Television, 2026-09-17.
5. Cómo cambia la inteligencia artificial la industria de la música. Jueves de Ciencia, Fundación UNAM and TV UNAM, Television, 2026-09-10.
6. Cómo cambia la inteligencia artificial el fútbol. Jueves de Ciencia, Fundación UNAM and TV UNAM, Television, 2026-09-03.
7. Autenticidad y redes sociales. ¿Son compatibles? Fundación UNAM, Television, 2026-05-28.
8. ¿Las redes sociales son una amenaza para la democracia? Fundación UNAM, Television, 2026-05-21.
9. ¿Pueden los chatbots sustituir a los psicoterapeutas? Fundación UNAM, Television, 2025-09-18.
10. Avatares: las nuevas estrellas. Fundación UNAM, Television, 2025-08-21.

11. De la IA a la toma de decisiones participativa: aplicación de optimización multi-objetivo a movilidad urbana sostenible. Consorcio Universidades por la Ciencia, Talk, 2025-08-19.
12. Tomando decisiones con aprendizaje por refuerzo. Guided visit for students of CCH-Azcapotzalco, Conference, 2025-05-13.
13. Regulación de la inteligencia artificial. Fundación UNAM, Television, 2025-03-27.
14. Chips de inteligencia artificial y el cerebro. Fundación UNAM, Television, 2025-03-20.
15. ¿Qué implica el arte generado por inteligencia artificial para los artistas? Fundación UNAM, Television, 2025-03-13.
16. Tomando decisiones con aprendizaje por refuerzo. Instituto Tecnológico Superior de San Martín Texmelucan, Conference, 2025-03-24.
17. Motores de búsqueda con inteligencia artificial. Fundación UNAM, Television, 2025-02-27.
18. Inteligencia artificial y las relaciones sentimentales. Fundación UNAM, Television, 2025-02-20.
19. Imágenes generadas por inteligencia artificial. Fundación UNAM, Television, 2025-02-13.
20. Inteligencia artificial hoy y en el futuro. Fundación UNAM, Television, 2025-02-06.
21. Un poco de inteligencia artificial y emprendimiento. Sociedad de Innovación, Ciencia y Emprendimiento, Conference, 2025-03-11.
22. Inteligencia artificial generativa y el emprendimiento científico. Sociedad de Innovación, Ciencia y Emprendimiento, Workshop, 2025-03-27.
23. Tanque de Pensamiento en Ciencia de Datos 2025. IIMAS, UNAM, Academic lead, 2025-01-22.
24. AlphaZero: integración de búsqueda simbólica y redes neuronales. Academia Mexicana de Ciencias, Workshop, 2024-11-11.
25. Un poco de aprendizaje por refuerzo y sus aplicaciones. Escuela de Estadística, Matemática y Ciencia de Datos en Memoria del Dr. Mario Alberto Díaz Torres, IIMAS UNAM, Conference, 2024-10-17.
26. Introducción al aprendizaje por refuerzo evolutivo. Encuentro Nacional de Computación 2024 (ENC 2024), Tutorial, 2024-09-24.
27. Optimización multi-objetivo y sus aplicaciones en aprendizaje máquina. Unidad Académica del IIMAS en el Estado de Yucatán, Conference, 2023.
28. Seminario de colaboración estudiantes-investigadores en ciencia de datos. IIMAS, UNAM, Conference, 2022-08-10.

29. La vinculación entre academia e industria y otros animales fantásticos. Primer encuentro de alumnos de la Licenciatura en Ciencia de Datos, UNAM, Conference, 2022-06-14.
30. Optimización de estrategias de manejo para vehículos autónomos mediante algoritmos multi-objetivo bajo incertidumbre. Diálogos IIMAS Ciencia de Datos, UNAM, Conference, 2021.
31. Posgrado: la puerta a las oportunidades. NEO 2021, Conference, 2021.
32. Posgrado: la puerta a las oportunidades. Instituto Tecnológico de Tepic, Conference, 2020.
33. Set-oriented methods for multi-objective optimization. Journal Club, Department of Engineering, University of Oxford, Conference, 2018.
34. Cell mapping methods for multi-objective optimization. NEO 2018, Mexico City, Conference, 2018.
35. Posgrado: la puerta a las oportunidades. Centro de Investigación en Inteligencia Artificial, Universidad Veracruzana, Conference, 2018.
36. Optimización multi-objetivo. Universidad Justo Sierra, Conference, 2013.

10 Academic Management and Institutional Service

- Head of the BSc in Data Science, UNAM (2026–present).
- Coordinator of the Applied Artificial Intelligence Diploma Program, General Directorate of Computing and Information and Communication Technologies (DGTIC), UNAM (2024–present).
- Representative of AI Area Advisors in the Graduate Program in Computer Science and Engineering (PCIC), UNAM (2025–present).
- Member of the Internal Gender Equality Commission, IIMAS–UNAM (2026–present).
- Member of the Collaboration and External Engagement Committee, IIMAS–UNAM (2024–present).

11 Participation in Conferences and Workshops

- CLAIO - Latin-Iberian-American Conference on Operations Research (2024)
- CSMIO - Conference of the Mexican Society of Operations Research (2016, 2025)
- Dagstuhl Seminar 16412 - Automated Algorithm Selection and Configuration (2017)

- EMO - International Conference on Evolutionary Multi-Criterion Optimization (2025)
- ENC - National Computing Meeting, Mexican Society of Computer Science (2024)
- EURO - European Conference on Operational Research (2019)
- EUROPT - Workshop on Advances in Continuous Optimization (2025)
- EVOLVE - A Bridge between Probability, Set Oriented Numerics, and Evolutionary Computing (2013–2014)
- GECCO - Genetic and Evolutionary Computation Conference (2019–2020, 2022–2025)
- MCDM - International Conference on Multiple Criteria Decision Making (2019, 2022, 2024)
- NEO - Numerical and Evolutionary Optimization (2013–2018, 2020–2022, 2024–2025)
- WCCI - World Congress on Computational Intelligence (2022)

12 Editorial Activities

12.1 Editorial boards and committees

- Associate Editor, Engineering Optimization (2026–present).
- Track Chair, Evolutionary Multiobjective Optimization track, GECCO 2026.

12.2 Peer review

Verifiable on ORCID: 0000-0003-4303-6866. 90 reviews for 15 journals and research projects (checked on 11/02/2026).

- Algorithms (2)
- Applied Soft Computing (2)
- Biomimetics (2)
- Engineering Optimization (15)
- European Journal of Operational Research (1)
- IEEE Transactions on Emerging Topics in Computational Intelligence (5)
- IEEE Transactions on Evolutionary Computation (31)
- IEEE Transactions on Fuzzy Systems (2)
- IEEE Transactions on Systems, Man, and Cybernetics (1)

- International Journal of Dynamics and Control (2)
- Journal of Heuristics (1)
- Journal of Multi-Criteria Decision Analysis (1)
- Mathematical and Computational Applications (6)
- Mathematics (10)
- Swarm and Evolutionary Computation (5)
- Symmetry (4)